# Shivam Tiwari

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## Summary

Senior Mobile Engineer Lead with **6+ years** architecting and delivering **high-performance, scalable, maintainable mobile applications** across **Android and iOS platforms** at scale (**180M+ MAU**). Deep expertise in **Native Android (Kotlin/Java)** and **React Native**, with **native module development** in Kotlin (Android) and Swift (iOS) — spanning the full **mobile development lifecycle** from architecture through release processes. Proven technical leader driving **technical decision-making**, promoting **engineering best practices**, mentoring engineers, and owning CI/CD pipelines across high-growth product companies. Collaborated closely with **product managers, designers, and backend teams** to ship user-friendly, high-quality features. Background in ML/Deep Learning research; applies on-device ML and **AI-assisted workflows** in production.

## Metrics Snapshot

- **53% faster CI/local builds** (30 min → 14 min) — Gradle optimizations, dependency caching, and modularization at Dream11
- **80% reduction in UI jank and re-rendering** across high-traffic Android chatroom screens at ShareChat
- **15MB+ APK size reduction** via Dynamic Feature Module (DFM) migration of FFMPEG and Agora at ShareChat
- **CriQ built 0→1**: Full-stack Android + iOS + React Native product with CI/CD, OTA updates, and multi-market release pipelines
- **App-wide framework owner**: Intervention Framework (ShareChat) — lifecycle-aware overlay orchestration system adopted company-wide
- **AI workflows in production**: Agentic PR review agents, test-generation pipelines, and on-device ML inference for real-time recommendations

## Experience

**Dream11** — Mumbai, India
*Senior Software Engineer – Mobile Lead* — *Dec 2024 – Present*

- **Led the 0→1 development of CriQ** across **Android** and **iOS** via **React Native**, backend services, CI/CD, and release pipelines end-to-end — supporting separate builds for Indian and international markets with social sign-in.
- **Reduced CI and local build time by 53%** (30 min → 14 min) via Gradle optimizations, dependency caching, and modularization; established **GitHub Actions** pipelines for automated releases and **CodePush** OTA updates.
- Built **native modules in Kotlin (Android) and Swift (iOS)** bridging on-device ML inference for real-time contest recommendations; integrated **Speech-to-Text and Text-to-Speech** dual-engine architecture in CriQ and the Dream11 flagship app — enabling conversational and accessibility features.
- Built **AI engineering workflows** in production: agentic automated PR reviews and unit-test generation pipelines, accelerating mobile team velocity and code quality.
- **Mentored junior and mid-level engineers** through structured code reviews, technical 1:1s, and pairing sessions; promoted **engineering best practices** across the full **mobile development lifecycle**.
- Collaborated closely with **product managers, designers, and backend teams** to deliver high-quality, **high-performance** mobile features — leading architecture and design reviews for scalability, modularity, and long-term codebase health.
- Drove **technical decision-making** across **Android**, React Native, iOS, and backend — identifying and resolving **production issues and performance bottlenecks**, and influencing engineering direction across multiple cross-functional initiatives.

**ShareChat & Moj** Bangalore, India
*Senior Software Engineer – Mobile* *Oct 2021 – Dec 2024*

- **Architect of the Intervention Framework** — a lifecycle-aware, app-wide system managing all overlays (popups, tooltips, snackbars) across app surfaces; used **Kotlin Coroutines Actor**, **MVI**, and **Dagger-Hilt** for queue orchestration and remote-controlled display logic; adopted company-wide for platform **reliability** and UX consistency.
- **Optimized high-traffic chatroom screens** in Android, reducing UI jank and unnecessary re-rendering by **80%** through layout hierarchy flattening, RecyclerView diffing strategy improvements, and Jetpack Compose recomposition audits — significantly improving scroll performance and frame stability.
- **Drove sustained crash-free rate improvements** across chat and chatroom modules through systematic ANR and crash triage, memory profiling, and targeted fixes — contributing to platform stability and reliability.
- Built and shipped **monetization features for audio/video livestreaming** using **Firebase Firestore** and **Agora RTC**, directly impacting revenue.
- Migrated high-footprint modules like FFMPEG and Agora to **Dynamic Feature Modules (DFMs)**, reducing app install size by **15MB+** and improving first-install conversion rate.
- Delivered production UIs in **Jetpack Compose** and **XML** across **MVVM / MVI / MVP**; implemented **testing strategies** with UI automation suites in **Maestro** and unit tests with **JUnit / Mockk**.

**Toppr / BYJU's** Hyderabad, India
*Software Engineer – Android* *Aug 2020 – Oct 2021*

- **Delivered BYJU's Future School from zero to launch in one month** with a team of 10 — animated educational videos, interactive questions, and gamified learning on an aggressive timeline.
- Co-developed **Toppr School OS** (team of 4), a full virtual schooling platform; owned Android architecture using **MVVM**, **Dagger-Hilt**, and XML. **Led the Android sub-team** alongside development responsibilities.

## Technical Skills

**Android**: Kotlin, Java, Jetpack Compose, XML, MVVM / MVI / MVP, Coroutines & Flow, Dagger-Hilt, Room, Dynamic Feature Modules, WorkManager, Paging3

**React Native**: React Native, JSI / New Architecture, Reanimated 3, FlashList, Expo, CodePush, TypeScript, React.js

**iOS**: Swift, iOS SDK, native module development

**Testing**: JUnit, Mockito, Mockk, Maestro (UI Automation)

**Backend & Web**: Node.js, SQL, RESTful APIs, Proto3/gRPC

**AI / ML**: On-device ML, Agentic Workflows, Speech-to-Text, Text-to-Speech, Deep Learning, Cursor, Claude Code

**Infrastructure & Tools**: Git, GitHub Actions, Firebase, GCP, AWS, Agora RTC, Datadog, JIRA, Confluence

## Education

**Indian Institute of Information Technology (IIIT) Guwahati** Guwahati, India
*B.Tech in Computer Science* – CGPA: **8.68 / 10** *July 2016 – Dec 2020*

- Programming Club Head — organised intra/inter-institute coding competitions, ran problem-solving sessions.
- ACM-ICPC Regional 2018 — rank 61/128 teams. 4⋆ CodeChef rating. Runner-up, NITS-HACKS 2018.

## Internships & Early Career

**Indian Statistical Institute (ISI) Kolkata** Kolkata, India
*Research Intern – Deep Learning* *May 2018 – Jul 2018*

- Researched unsupervised representation learning using **Autoencoders**; explored latent space structure for feature extraction on sensor and image data under faculty supervision at one of India's premier research institutions.

**SJTech Solutions Pvt. Ltd.** Bhopal, India
*Part-time Data Scientist* *May 2019 – Jan 2020*

- Built and deployed computer vision models for **object detection & localisation** (garbage, hand-pump detection in web-scraped imagery), **face/gender recognition**, and **real-time drowsiness & motion detection** from live camera feeds using Python and deep learning frameworks.

## Research Publications (Full list on Google Scholar)

**Enhanced Annotation Framework for Activity Recognition via Change Point Detection** *IIIT Guwahati*

- Designed an S-CPD (Segmented Change Point Detection) framework to learn activity change points from unannotated sensor datasets and generate labels automatically.

**Semi-supervised Subject Recognition in Low-Modal Sensor Data** *IIIT Guwahati*

- Built two semi-supervised frameworks for subject recognition across ubiquitous and vision-sensor datasets under low-modality, scarce-label conditions.

**Behavior Analysis via Routine Cluster Discovery in Ubiquitous Sensor Data** *IIIT Guwahati*

- Modelled latent feature distributions for user-specific behavioural characterisation from ubiquitous sensor data.

**Deep Unsupervised Methods for Behavior Analysis in Ubiquitous Sensor Data** *IIIT Guwahati*

- Developed a self-supervised approach for latent behavioural feature learning; demonstrated improved user modelling without labelled data.

**Hyperspectral Image Classification using Semi-supervised Deep Learning**

- Built a deep learning pipeline using clustering-generated pseudo-labels for hyperspectral image classification, reducing dependence on manual ground-truth annotation.